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# Famous Astronomers: Carl Sagan

The Cosmic Communicator

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## Introduction

Carl Sagan, who lived from 1934 to 1996, was an American astronomer and planetary scientist who became the most recognizable face of science across the middle fifty years of the twentieth century. He was a serious researcher with real discoveries to his name: he worked out why Venus is a furnace, explained the seasonal darkening of Mars, predicted the orange haze of Titan, and helped design the first physical messages our species ever flung toward the stars. He was also a teacher, an author, and, with the 1980 television series Cosmos, a broadcaster whose voice reached hundreds of millions of people in dozens of countries.

This book follows both halves of that life. The first half belongs to the working scientist: the Chicago graduate student who wrote a doctoral thesis on the origin of life under a future Nobel laureate, the young Harvard researcher who calculated the greenhouse oven hidden under the clouds of Venus, the Cornell professor who read the first close-up pictures of Mars and Jupiter like anyone else reads a page. The second half belongs to the communicator: the man who turned planetary science into a planetary story, who argued that extraordinary claims require extraordinary evidence, and who insisted that science is a candle in the dark rather than a pile of facts.

The two halves were not in tension. Sagan believed, from childhood onward, that wonder and skepticism are the same instinct applied in different directions, and his career was a long demonstration of the point. The questions he asked as a researcher, whether life could survive on Mars, whether the early Earth was lit by a dim red sun, whether distant civilizations could exist and how we might ever know, were exactly the questions he asked on television. He simply refused to accept that the general public had no right to hear the answers.

The chapters that follow move through his life roughly in order: a Brooklyn childhood seized by astronomy at the age of five, an education at the University of Chicago under some of the greats of the age, the Harvard and Cornell decades that produced his science, the spacecraft missions he helped shape, the message projects that made him famous far beyond astronomy, the television career that made him a household name, the battles against pseudoscience that consumed some of his late energy, and the exoplanet-era legacy that has outlived him. A closing set of chapters collects the human details, the open questions, and the meaning of a life spent trying to give the whole species a sense of its own address in the universe.

One note on how to read what follows. Sagan's name is attached to catchphrases, billions and billions, the pale blue dot, extraordinary claims, and the temptation is to file him as a media figure who also held a professorship. The chapters ahead are an argument against that filing, made with the evidence: the greenhouse on Venus, the dust storms of Mars, the orange moon Titan, the nuclear winter calculations, the Voyager records, the exoplanet strategy that his old papers sketched before there were exoplanets. The reader who wants only the man and the jokes can jump to the chapter of trivia near the end, and it will be here when they want the rest.

## Chapter 1: The Discovery of Carl Sagan

Carl Edward Sagan was born on November 9, 1934, in the Bensonhurst section of Brooklyn, New York. His mother, Rachel Gruber, was a New Yorker from a poor family; his father, Samuel Sagan, was a garment worker born in what is now Ukraine, then part of the Russian Empire. The Sagans were Reform Jews, modest and not especially observant, living in a small apartment through the final years of the Great Depression. Samuel worked his way up from ushering in a movie theater to running the shop floor; Rachel kept an intense, sometimes anxious intellectual ambition that she channeled largely into her only son.

Biographers, above all Keay Davidson, whose 1999 life of Sagan remains the standard biography, have noted the division of labor in the boy's formation. From his mother Sagan traced his analytical drive, the habit of examining every claim. From his father he traced his sense of wonder. Samuel Sagan gave apples to poor neighbors and mediated labor disputes in the garment trade, and he answered his son's endless questions patiently. Sagan himself later described his parents as people who knew almost nothing about science but who introduced him, simultaneously, to skepticism and to wonder.

Two encounters in his early childhood fixed his course. The first came from a library card. At roughly age five or six, by his own account, Sagan asked a librarian for a book about stars. She first brought him a book of Hollywood portraits, a confusion that amused him for the rest of his life, and then found the right kind of book. In it he read that the stars are suns, very far away. The scale of the universe opened up to him at once, and he later said he never recovered.

The second encounter was the 1939 New York World's Fair, which his parents took him to at age four. He remembered the Futurama exhibit, with its miniature highways and gleaming cities of the future, and he remembered a time capsule buried at the fairgrounds, a message sealed in the present for people of the distant future. The idea that you might deliberately send a message across an abyss of time lodged in him and resurfaced decades later in the plaque he helped design for Pioneer 10 and the golden record carried by the two Voyager spacecraft. Sagan loved to point out this symmetry: his first remembered wonder was a time capsule, and one of his last great projects was exactly that.

The war years made the family anxious for relatives in Europe, some of whom were caught in the Holocaust, though the child was shielded from the details. In 1947 the family moved across the bay to Rahway, New Jersey. Sagan's interests were already unshakable. He kept a friend's father in the neighborhood, an amateur astronomer, on call for questions; he read constantly; he wrote a school essay arguing that human contact with a vastly more advanced civilization might go as badly for us as first contact between Europeans and Native Americans had gone for the Native Americans, a thought he carried intact into his adult writing about extraterrestrial life.

In high school he led the astronomy club and, in a move that says much about his confidence, wrote to the director of Harvard Observatory asking to be shown around. He was invited, met the director, and left convinced that the profession was his. At sixteen he won a scholarship and enrolled at the University of Chicago, which then had a policy of admitting young students who wanted a genuinely liberal education. Brooklyn, the apples, the library book, and the fair: by 1951 the raw material of the later career was all in place, waiting for the training.

A few more details from the early years belong here, because they show how complete the pattern was before he ever trained as a scientist. The books that grabbed him after the library book were the Mars novels of Edgar Rice Burroughs and the other planetary romances of the pulp era, and Sagan later said frankly that the stories were nonsense but the hunger they fed was real: they made Mars a place, and places can be visited and studied. He kept the affection for science fiction his whole life, and it repaid him twice, first as an audience for his own novel and later as a genre he helped keep honest.

His reading ran wide, not just deep. Alongside astronomy he absorbed paleontology and devoured everything he could find about evolution; in the standard anecdote, at age five he was already asking an adult what a billion years really meant and being told that nobody knew, and finding the answer unacceptable. In Rahway he and a friend acquired a small telescope or at least borrowed one often enough for the visits to count, and he learned the constellations from the northern New Jersey sky, which even then was washed out by city light, a fact that irritated him and later made him a quiet campaigner against light pollution.

There was also a family influence that biographers weight heavily: his sister, Carol. The two agreed in adulthood that the household had a particular emotional weather, a mother whose anxiety could fill a room, and that Carl's early outward calm was partly built over it. The point matters because the mature Sagan was famous for composure on camera, and that composure was constructed early, out of materials like these.

He graduated high school at sixteen, having already won the citywide science awards and essay contests that made Chicago's admissions office take notice. The university's great-books-influenced curriculum meant he was made to read Plato and Darwin and Freud alongside the physics, and he credited that breadth for the writing style that later sold millions of books. By the time he left New Jersey, the plan was fixed: he would be an astronomer, he would study the planets, and he would explain what he found to anyone who would listen. The rest of this book is what happened to that plan.

A last childhood fact, kept to the end because it explains the adult's moral seriousness. The Sagans were working-class Brooklyn, and the Depression and the war taught the household economy and dread at the same time; Sagan's later writing about poverty, about the waste of human talent in bad systems, and about the responsibility of the lucky, is autobiographical at the root. He told interviewers more than once that the two memories he could not shake were the fair's time capsule, built by people who assumed there would be a future to open it, and the war news, which arrived in the same years, and that his whole career sat between those two poles: the assumption of a future, and the need to make sure there is one. That is the frame for everything the chapters ahead will show.